Journal of Intelligent & Robotic Systems (2021) 101: 31
https://doi.org/10.1007/s10846-020-01300-2

Design and Development of a High-Speed UAS for beyond Visual Line-of-Sight Operations

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Received: 21 February 2019 / Accepted: 14 December 2020 / Published online: 22 January 2021

Abstract

This paper provides details on best practices for vehicle design for BVLOS missions, with a focus on robust control links and real-time health monitoring.The development work presented herein focuses on a specific mission to set the speed and distance world record for a fully autonomous vehicle, requiring development of analytic tools for estimating vehicle range and endurance, as well as models for predicting command and control link integrity while operating BVLOS. Range and endurance estimates are developed using a novel real-time system identification algorithm onboard the vehicle for aerodynamic parameter estimation, and propulsion system characterization. Flight-testing methods and results are presented demonstrating the capabilities of these algorithms, with results from a unique test-case involving an unintended fault during the record flight. In addition to the vehicle performance assessment, Friis-based models are introduced to provide a pseudo real-time communications link range assessment. The predictive communications range models were refined using specific antenna and system characterization data measured in a compact range facility resulting in a detailed link budget useful for addressing BVLOS command and control requirements. The combination of aircraft and radio-frequency performance prediction models with supporting flight-testing techniques applied to small unmanned aerial systems provides critical insight into the design of future vehicles, and validation of existing systems tasked with performing BVLOS missions.

Keywords UAS · Beyond visual line of sight · Flight testing · Aircraft systems design

1 Introduction

Enabling safe beyond visual line-of-sight (BVLOS) operations in the national airspace system (NAS) for small unmanned aircraft systems (UAS) is a key technical hurdle for wide-scale commercial use of these vehicles [1–4]. Such operations are currently prohibited under Federal Aviation Administration (FAA) Part 107 rules which govern operations of UAS in the NAS [5]. This ruleset restricts operations to flights below 400 ft., unaided visual line of sight, and daytime visual-flight-rules operations only. While these restrictions allow for limited operations in the NAS, most proposed package delivery and other routine flights would not fit within this restrictive operating envelope. The technical hurdles for demonstrating safe operations beyond the limits imposed by Part 107 primarily fall on measuring or demonstrating the reliability and capabilities of the vehicle. In this paper, we detail the key design guidelines and flight-testing procedures for a high-speed UAS targeted at demonstrating BVLOS capabilities. To demonstrate the design and flight-test assessment process for a BVLOS capable vehicle, a choice was made to pursue a mission which would qualify as a world record for an autonomous aerial vehicle. The specific flight path is dictated by the record governing body, the Fédération Aéronautique Internationale (FAI) and is detailed in subsequent sections. Completing this mission requires careful design and planning to ensure the capabilities of the vehicle and its subsystems meet the stringent safety requirements for extended operations outside of direct visual line of sight.

Range and performance estimation for UAS has primarily focused on electric propulsion systems, given their low-cost and relatively high performance on small platforms [6, 7]. However, enabling high-speed flight with usable endurance and payload capabilities still favors gas-turbine powerplants. Tavakolopour-Saleh et al. demonstrated a system

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identification approach to developing a performance model for a small turbojet engine [8]. These models were demonstrated in static ground testing, but were not extended to flight, where significant deviations from the reported dynamics exist due to the open-loop functionality of the turbine's built-in control system. Similarly, Jiali and Yang proposed a non-linear system dynamics model to capture the open-loop response of the small-scale JetCat powerplants, though the tests were limited to ground runs. Grannan et al. extended ground testing to incorporate an altitude chamber to investigate the effects of changing static pressure on engine performance and noted additional non-linear behaviors likely tied to the lack of barometric compensation in the engine control unit. Flight-testing of small turbojets representative of those used in this work has been conducted by Gu [9]. In their work, systems identification techniques were applied to estimating thrust and vehicle aerodynamic characteristics in steady, level flight, though no specific information was presented on the efficiency of the turbine system in the form of thrust specific fuel consumption (TSFC) or its variation across the vehicle performance envelope. This work builds on previous characterization studies for small turbojets and presents methods for estimating their performance in flight and resulting impact on range and endurance informed by vehicle dynamics and validated through flight-testing.

Command-and-control (C2) link integrity continues to be a primary concern for integration into the NAS. Numerous working groups and standards agencies have convened to address specific issues involving predicting spectrum availability and reliability, such as those conducted by NASA, RTCA, ASTM, and 3GPP. The work presented by Ivancic et al. presents a complex and accurate framework for assessing network coverage for ISM and protected bands including 3/4G cellular networks [10]. However, while these models perform well with numerous data sources distributed throughout a given environment, they require external data not available to a remote UAS for forecasting link integrity. Stansbury et al. provided a comprehensive overview of spectrum use and availability for BVLOS missions, but indicates that predictive tools are lacking for terrestrial line-of-sight range estimation [11]. In contrast, the work presented here provides a lightweight range prediction tool which may be incorporated into even the most basic autopilot or flight management systems currently fielded in small UAS, without the need for external data sources.

The following design guidelines for enabling BVLOS capabilities will begin with a description of the intended mission profile for the demonstration flight. Within this discussion, a basic vehicle performance model and mission segment analysis is proposed. Using the mission constraints as a framework, a detailed aircraft performance assessment is presented, and specific details for the integration of the C2 links are developed. These systems-level characterizations are then validated through flight test, demonstrating the applicability of the methods in characterizing the range and communications capabilities for BVLOS operations.

2 Aircraft Mission and Systems Overview

The mission overview and baseline vehicle configuration are detailed in this section and are further refined through flight tests in subsequent sections. Specific details of the flight management computer and inertial navigation system which enable the detailed flight performance estimates are provided in section 4.

2.1 Mission Overview

The mission parameters follow directly from the requirements set forth from the FAI unmanned aircraft speed and distance rules [12]. The minimum course length for an unmanned speed record requires out-and-back legs of between 8.1 nm (15 km) and 13.5 nm (25 km) distance, with additional 2.7 nm (5 km) stabilized approaches (entry regions) preceding the high-speed legs. The minimum distance of 8.1 nm was selected for this mission, as illustrated in Figs. 1 and 2. The lateral extents of the course are ±0.54 nm (±1 km) on either side of the high-speed corridor, and ± 330 ft. (±100 m) in altitude. Distance records are declared by identifying individual legs, which for the purposes of this work were the sum of the speed legs and three of the approaches of the course (the approach at the end of the course need not be flown for a record). In this work, an operational range of approximately 13.5 nm was identified over open water at altitudes not to exceed 1000 ft. AGL. The concept of operations (CONOPS) and operational area were developed in close coordination with the FAA using a special exemption available to public institutions, the Certificate of Waiver or Authorization (COA). A hazard assessment was conducted to ensure minimal risk to other aircraft, or to people and property on the ground. A waiverable flight operations area located over Lake Erie was identified as the optimal location for the record flight. The area abuts Kelleys Island which has a local airfield with a paved runway approximately 2200 ft. long which terminates directly into the lake on the eastern edge of runway 9. This base of operations provides clear visual and C2 paths for ensuring positive vehicle control during the flight.

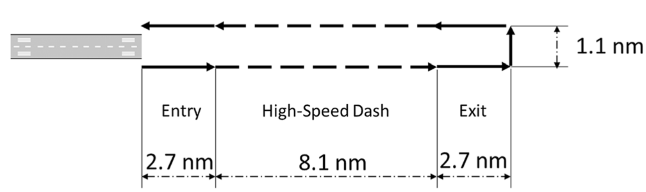


Fig. 1 FAI course dimensions for velocity and distance mission

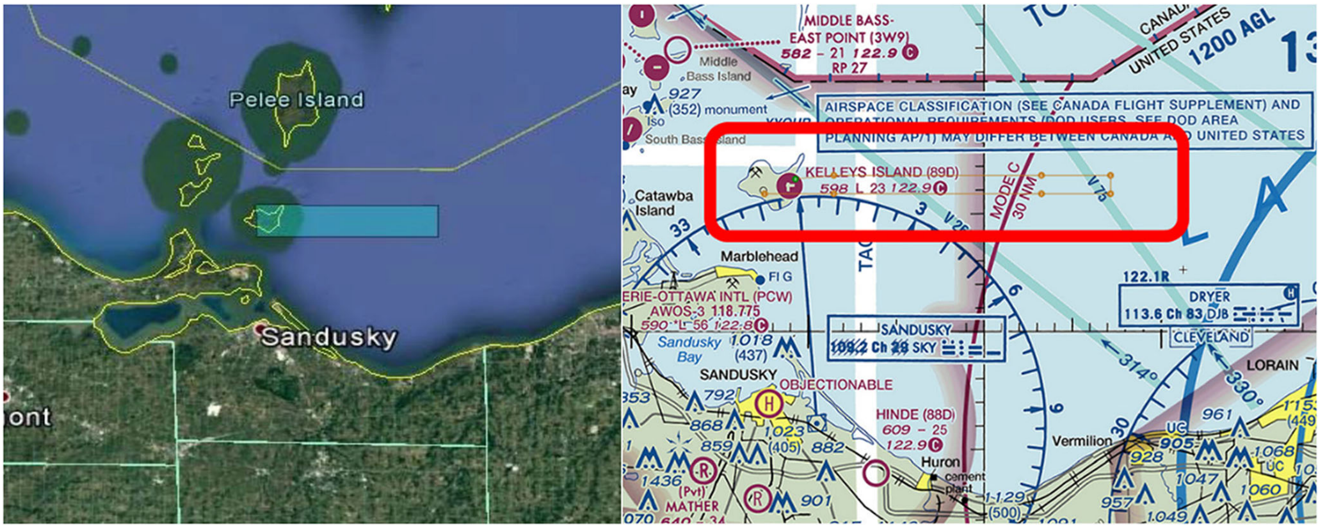


Fig. 2 Approved operational area via Certificate of Waiver or Authorization. Google Earth (left), FAA sectional (right)

2.2 Aircraft Systems Overview

The specific focus of this work is enabling BVLOS capabilities for small UAS, which typically enforces an upper bound on the gross takeoff weight of 50 lbs. This corresponds to the FAI U-2.b class (5–50 kg), which also extends the minimum out-and-back course length, necessitating true BVLOS capabilities. To meet the course requirements, a decision was made early in the program to leverage an existing commercial-off-the-shelf (COTS) airframe to both demonstrate the procedures for assessing the utility and performance of an existing airframe, and as a cost control measure. Numerous turbojet powered airframes are available from the hobby community, and after down selecting based on estimated top speed, and internal payload volume, the SkyMaster Avanti XXL airframe was chosen. This vehicle utilizes a turbojet powerplant (JetCat P-180RXi) which provides substantial thrust at the cost of exceptionally high fuel-consumption. This combination netted the highest overall speed for the intended course when compared to similarly sized vehicle powered by internal combustion engines or electric motors with propellers. The basic vehicle configuration and relevant scale parameters are given in Fig. 3 and Table 1.

Table 1 Vehicle dimensions and flight characteristics

Metric		
Weight (gross)	70	lb
Weight (empty)	40	lb
Span (wing)	97.6	in
Area (wing)	12.67	ft^2
Length (fuselage)	124	in
Thrust (Max takeoff)	40	lb
Stall speed	60	kts

Endurance and maximum velocity were critical constraints in vehicle selection. However, internal volume was also important to ensure adequate space for integration of the various C2 and flight management computer (FMC) systems required for the flight. A high-level functional diagram for the onboard systems is shown in Fig. 4 with details on the connections between components and associated protocols. The FMC was created entirely in house at the Ohio State University and is described in Section 2.3. These components are the core of the guidance and control system and were designed to flexibly integrate with other sources of vehicle state information and the redundant C2 links (Fig. 5, with antenna placements shown to scale). A custom inertial navigation system (INS) interfaces with the FMC to provide the state estimates needed for vehicle control. The ground control station (GCS) uses a highly modified instantiation of QGroundControl to provide enhanced situation awareness to the ground crew from airborne and ground-based ADS-B sensors. Communications between the GCS and FMC via the multiple redundant communications links on the vehicle use the open-source Mavlink protocol. Mavlink has emerged as a pseudo-standard for open-

Fig. 3 High-speed turbojet research aircraft, the Avanti

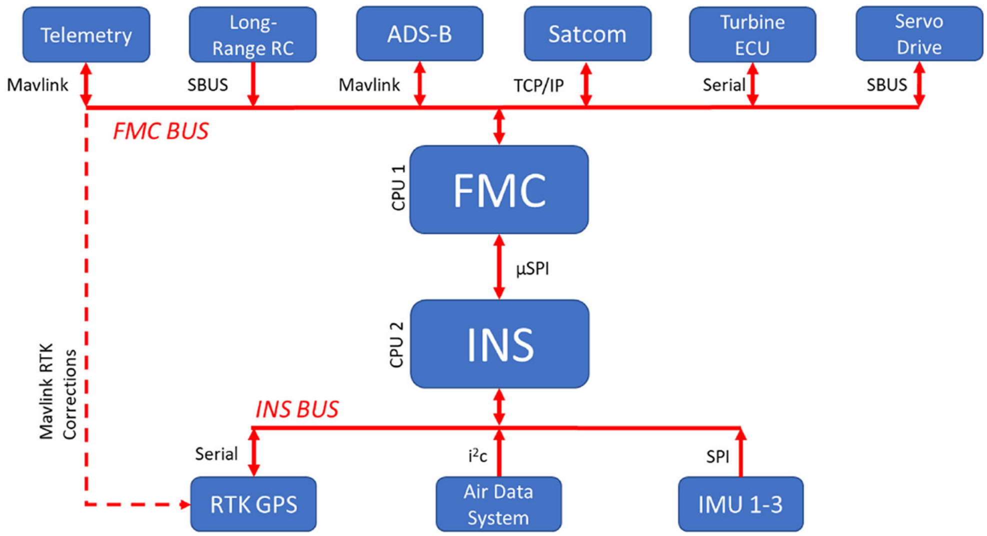


Fig. 4 Vehicle avionics diagram. Arrows correspond to information busses with associated protocols

source flight vehicles and is an efficient byte-ordered protocol used to transmit state and control commands with relatively low bandwidth requirements. Ground-based directional antennas for the control links were mounted on steerable gimbal to enable tracking of the vehicle in flight.

The Avanti employs triple-redundant physical-layer control links, with terrestrial links on 900 MHz and 433 MHz, and geosynchronous satellite communications on 1.53 GHz. The primary control link relies upon a 1-W, 915 MHz transceiver for telemetry with diversity monopole antennas and copper ground planes located on the bottom of the left and right wingtips. Backup long-range vehicle control is provided by a 433 MHz long-range transceiver and ¼ wave dipole antenna affixed inside the vertical stabilizer. This system allows for direct control between a standard ground-based remote-control (RC) transmitter while providing limited telemetry information up to 21.6 nm (40 km) from the ground base of operations. A dedicated ViaSat satellite transceiver is installed on the aircraft to demonstrate techniques for integration of high-latency, high-reliability redundant communications links. This link provides coverage over the CONtinental United States (CONUS) and can enable missions where direct RF link of sight is unavailable due to terrain or obstacle masking. For this mission, this system was used as a back-up link for issuing abort or return to base commands to the vehicle. The antenna used for this system is highly directional and requires an unobstructed line-of-sight view of the sky. It is mounted in the canopy housing and is selectively shielded to minimize the effects of EMI. Telemetry information provided by the satellite link is displayed on the GCS using an identical ground based ViaSat receiver.

A 1090/978 Automatic Dependent Surveillance-Broadcast (ADS-B) system is used onboard the vehicle to track cooperative targets and is used as a redundant means of tracking the Avanti in flight. The antenna for this system is mounted inside the fiberglass fuselage, behind the carbon fiber supports for the landing gear, and immediately ahead of the long-range fuel tanks. Data from the ADS-B transceiver is fed into the FMC for real-time autonomous traffic deconfliction capability, and in parallel is downlinked to the GCS for display to the operator. A 600 mW, 5.8 GHz video link is also used for data recording and ground operations, but not for direct flight control. This system is mounted in the forward part of the canopy to provide an unobstructed view.

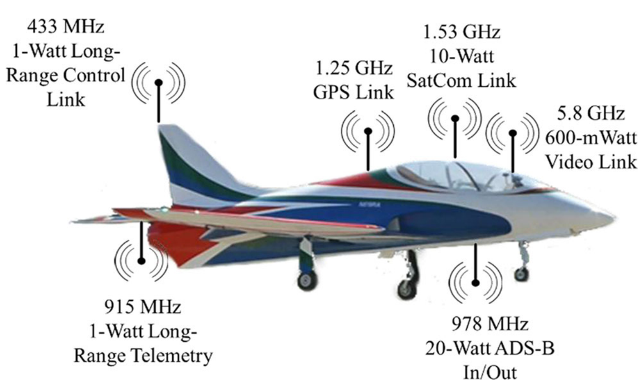


Fig. 5 Onboard C2 systems shown with approximate physical installation locations

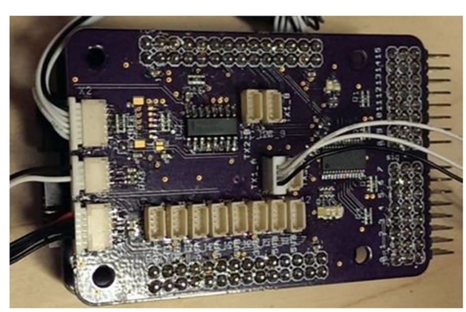
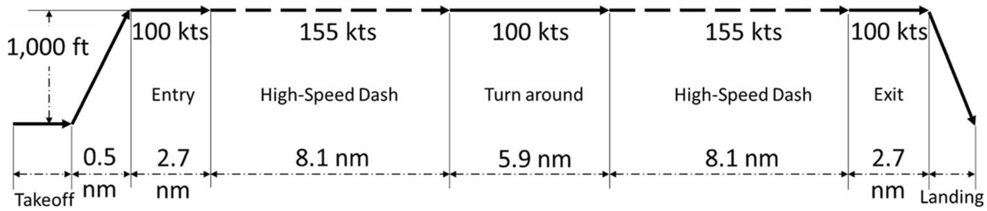


Fig. 6 Flight management computer and inertial navigation system

Fig. 7 Course dimensions and velocities



2.3 Navigation and Guidance System Details

The INS and FMC (Fig. 6) provided an assessment of inertial data quality on vehicle state estimation and control [7]. This system utilizes triple-redundant inertial measurement units (IMUs) with measurement uncertainties ranging from commercial grade to sub-tactical grade solid-state systems found in modern aircraft avionics. Aside from redundancy, the availability of sensor data with varying levels of uncertainty allows for assessment navigation performance with low-cost MEMs based IMUs typically found on small UAS. Data from the three IMUs are fused using an error-state Unscented Kalman Filter [13, 14]. The use of solid-state sensors necessitates a strap-down mechanization of the equations of motion, which have been specialized in this application by omitting higher-order transport rate terms not observable in the IMU data [15, 16]. However, the Coriolis terms typically omitted in flat-Earth navigation solutions are retained given the higher flight-speeds and longer distances associated with this vehicle. Unaided, the fused IMU data provides stable navigation solutions for in a static environment for approximately 5 min, with 95% velocity and position error bounds of approximately 1 in/s and 13 ft. Under dynamic motions, these errors are compounded by omission of higher-order dynamic terms such as those appearing in coning/sculling corrections outlined by Savage [15, 16]. To address these inherent instabilities, external aiding sensors are employed including a Real-Time Kinematic (RTK) GPS system, which provides accurate differential measurements of the vehicle relative to a fixed base station. The specific system employed in this work uses the open-source RTKLIB software package with a survey-grade base station providing correction data to the GPS system onboard the vehicle. The corrected positional accuracy of these sensors is better than 2 in. at distances less than 20 km.

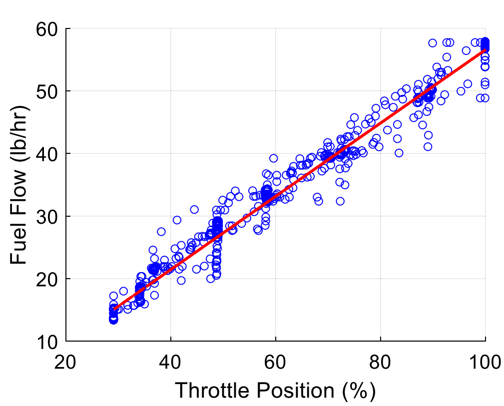


Fig. 8 Fuel flow-rate mapping during static ground test

Additional developments beyond the standard strap-down attitude formulation were adopted to facilitate flight at higher speeds with unknown aiding sensor latency. For example, the RTK GPS system provided position correction measurements with a mean accuracy of approximately 1-in in static conditions. However, the latency was randomly varying with a mean component of ~0.5 s with a standard deviation of ±0.1 s when sampled at 4 Hz. At a flight speed of 155 kts, the mean latency corresponds to a positional error of approximately 130 ft. Typical methods of addressing this error involve storing all IMU data between state vector corrections and back-propagating the correction vector [17, 18]. This approach requires significant computational overhead, and additional data storage requirements. Instead, a novel approach was adopted in which the delayed GPS signal was propagated forward using a 4th-order state reconstruction based on the relative change in vehicle state provided by the INS. This method implicitly assumes the relative state updates based on GPS message latency are accurate, but as shown by Khosravian et al., the methods are asymptotically stable under most flight conditions [19].

The flight management computer (FMC) developed for the Avanti incorporated many fault-tolerant design features to ensure safe operations and predictable behavior in the event of an inflight failure. This included direct closed-loop control of the JetCat engine via data provided by the turbine engine control unit (ECU). This system provides a real-time indication of turbine performance parameters including RPM, fuel flow, EGT, and a fuel totalizer. These measurements were augmented with an air data system which incorporated total and static pressures from a 5-hole probe located on the nose of the vehicle [20]. As an ultimate fail-safe mechanism, an independent servo control system was developed which electrically isolates the FMC in the event of a failure and would allow the ground-based PIC to take full manual control of the aircraft. This safety feature can only be initiated by the PIC using a redundant radio link to the vehicle and was tested in simulation to ensure adequate manual vehicle control for emergency maneuvering.

Table 2 Fuel burn weight fractions (asterisk data from Raymer [23])

Segment	W_f/W_o	*Planned Range*	*V*	*Assumed Lift/Drag*	TSFC (1/h)
Engine start/taxi/departure	0.970*	–	–	–	–
Climb	0.998	0.5 nm	100 kts	8	2.34
Gate 1	0.992	2.7 nm	100 kts	8	2.34
High-Speed 1	0.979	8.1 nm	155 kts	6	2.34
Gate 2, 3, & Turnaround	0.982	5.9 nm	100 kts	8	2.34
High-Speed 2	0.979	8.1 nm	155 kts	6	2.34
Gate 4	0.991	2.7 nm	100 kts	8	2.34
Landing	0.995*	–	–	–	–
Total	0.894				

3 Aircraft Range

Estimation of the range and endurance for small UAS operating BVLOS is of critical importance for ensuring safety of flight. Traditional range estimation tools focused primarily on vehicle fuel limitations must be augmented with explicit C2 range estimates to ensure reliable communications exist with the craft considering uncertainties in the terrain and propagation characteristics of the RF signals at low altitudes. Reliable methods for estimating vehicle performance for small UAS and C2 link availability are presented in this section.

3.1 Fuel Range Estimation

Classical range estimation techniques for turbojet powered aircraft have been well established. However, to date their applicability to small UAS vehicles has not been demonstrated. The methods presented here follow from the Breguet range equation and rely on segmenting the flight into regions with relatively constant velocity and associated fuel burn [21]. These phases are outlined in Fig. 7, and correspond to the mission segments outlined in section 2.1. The Breguet range equation for a constant velocity turbojet is given as,

$$\ln\left(\frac{W_o}{W_f}\right) = \frac{R \cdot TSFC}{V\left(\frac{L}{D}\right)}, \tag{1}$$

where the weight fraction, $\left({}^{W_o}/\,W_f\right)$ represents the fractional fuel burn of the particular mission segment, R is the length of the segment, *TSFC* is the thrust-specific fuel consumption, V is the flight speed, and *L/D* is the lift-to-drag ratio for the vehicle. The mission segments outlined in Fig. 7 provide the length of the individual elements and corresponding velocity based on initial flight test data. The TSFC data was estimated using ground-based static runs as shown in Fig. 8. The aerodynamic characteristics of the vehicle were estimated using drag approximations outlined in Phillips [21] and Hoerner [22], along with baseline estimates provided by computational panel methods such as XFLR5.

Table 2 lists the weight fraction estimates assuming a constant TSFC derived from the maximum measured thrust and fuel-flow rate. The validity of this assumption is addressed in following sections. With a maximum takeoff weight of 70 pounds, and a total estimated fuel weight fraction of 0.894, 7.42 pounds (1.09 gal) of Jet-A are required for this flight. To account for uncertainty in the fuel range estimates, a margin of safety of 2 was employed in the design. Also, planned fuel reserves of 1.0 gal (6.81 lbs., corresponding to 95% Jet-A and 5% turbine oil) were included to provide an additional 10 min of flight at 70% throttle. Thus, the total fuel carried on board was 21.7 lbs. (3.18 gal). The stock airframe included a 2-gal tank, so a custom-fabricated supplemental tank was installed in the fuselage directly forward of the main tank to achieve the required 3.18-gal capacity.

Table 3 Friis range and Fresnel radii estimates for Avanti aircraft

System	Frequency	P_t(W)	P_r (W)	G_t(dBi)	G_r(dBi)	Range estimate (*d,nm*)	Fresnel radius (*F, ft*)
Telemetry	915 MHz	1.0	1e-12	1.0	0.95	51	593
Control	433 MHz	1.0	3.16e-13	6.5	2.11	52	412
ADS-B	978 MHz	20.0	–	–	1.62	236	849
Video	5.8 GHz	0.6	3.16e-12	1.0	1.17	1.0	22

Fig. 9 Antenna characterization at using OSU's Compact Range Facility

3.2 RF Range Estimation

3.2.1 Compact Range Testing

Incorporation of multiple, high-power transceivers in relatively close proximity on a composite airframe presents a formidable challenge in limiting interference and establishing adequate gain patterns for the antennas and frequencies of interest. For this vehicle, transmission occurred in the UHF and SHF bands corresponding to the systems listed in Table 3. To characterize the range capabilities of each system, the vehicle was tested in OSU's Compact Range Facility at the Electro Science Laboratory (Fig. 9). During the characterization tests, the individual antenna elements were connected to low-power sources

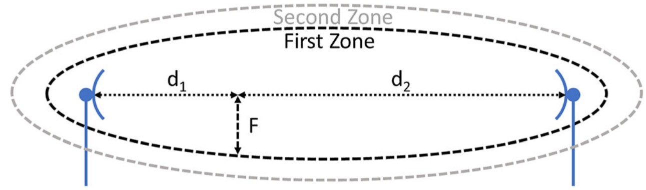
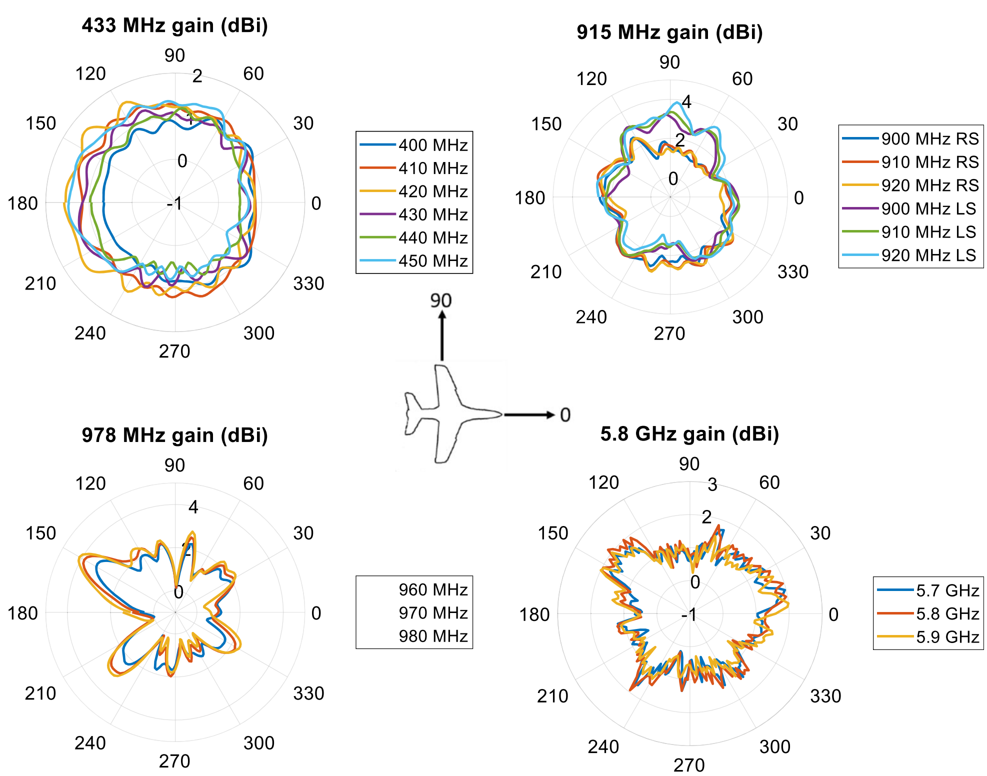


Fig. 11 Fresnel zone depiction

capable of providing excitation across the bandwidth of the radio under test. Received signal strength from a separate calibrated horn antenna system in the chamber allowed for assessment of the gain patterns and any signal attenuation created by the other radiating elements or airframe components.

The resulting radiation patterns for the individual systems are shown in Fig. 10. The 433 MHz system exhibits relatively isotropic gain, with a slight variation between 0 and 180 degrees. This is due to the location of the antenna inside the vertical tail of the aircraft, where there is more carbon support material in front of the antenna than on the sides/back. The 915 MHz antennas are a diversity pair mounted on bottom of the left and right wingtips. The gain variation between the two antennas is clearly visible with left antenna having a higher gain when viewed from 90 degrees, and the right antenna exhibiting similar behavior when viewed from 270 degrees. The left/right gain difference is due to the proximity of the fuselage effectively blocking reception from the far antenna. However, the diversity functionality built into the radio automatically selects the highest received power source, thus providing isotropic performance on a system level. The 978 MHz

Fig. 10 Azimuthal gain characteristics for the command and control telemetry systems



ADS-B plots show significant azimuthal attenuation with a maximum null of 2 dB occurring at 90 and 180 degrees. The cause of this attenuation is airframe interference from conductive composite supports (90 deg) and the fuel tank (behind the antenna, 180 deg) in close proximity (~5 cm). While this is not a flight-critical link, the anisotropic gain is generally offset by its power rating (20 W) and low duty cycle operation (1 packet/s). Of note, noise floor estimates were provided by the integrity monitors built into the radio systems. Once the received power falls below this level, the signal no longer is considered viable. When combined with the power output levels and gain statistics, this provides the final information necessary to quantify the predicted range of the system and will be discussed in the following section.

Significant time and resources were spent integrating the various antenna systems into the aircraft. The majority of this time involved repositioning the antenna systems inside the aircraft to minimize the gain versus azimuth distortions arising from interference with the various transmissive (fiberglass) and reflective (carbon fiber and metallic) materials used in the vehicle. This highlights a prime difficulty in integrating robust communications links into a new or existing small UAS and should be considered a prime design constraint for new vehicles. Additionally, many small UAS use composite materials for a large part of their substructure. While these materials offer many mechanical advantages for shape optimization and weight savings, the resulting airframes have little in the way of effective ground planes necessary for maximum RF efficiency. Coupled to this, the lack of suitable grounding planes can result in significant EMI and ground loop issues when high-powered systems are in close proximity to one another, which is a given for most small vehicle designs.

3.2.2 RF Range Modeling

The 433 and 915 MHz C2 radios provide measurements of real-time transmit and receive power (see Table 3). The Friis transmission relation provides the empirical underpinnings to relate transmitted and received power as a function of frequency and the gains of the respective antennas. The Friis relation provides an indication of expected RF signal power as a function of the distance between the transmitting and receiving antennas based on the medium through which they are transmitting (e.g., air or vacuum). The Friis relation is given as

$$d = \left(\frac{\lambda}{4\pi}\right)\sqrt{\left(\frac{P_t G_r G_t}{P_r}\right)}, \tag{2}$$

where $P_{t,\,r}$ are the transmit and receive power, respectively, $G_{t,\,r}$ are the isotropic gain of the transmit and receive antennas (discussed in the previous section), λ is the RF wavelength, and d is the distance between antennas [24, 25].

Here, the Friis relation is used with measured or estimated parameters to determine the anticipated maximum range, d, between the antennas. The initial system design utilized tabular gain factors for the individual antennas. Later, these approximations were replaced with direct measurements acquired using the compact range facility. Noise floor estimates are provided by the radios and represent the minimum received power for successful communication. Predicted range capabilities are shown in Table 3 for the mean gain values extracted from the compact range testing. The 433 MHz control link is primarily sending information from the GCS to the aircraft, making the transmit power and gain relative to the ground-based system. However, the telemetry link is bi-directional with a different ground-based antenna (directional) than the monopole antenna on the aircraft. The ground-based and aircraft telemetry radios are identical, with the same power output, bandwidth, and noise floors (this will be addressed in the subsequent testing section). In this case, the transmit subscript refers to the ground-based installation, and the receive subscript to the aircraft. Of note, the ADS-B system is a transmit-only installation onboard the aircraft with a predicted range in line with ADS-B standards.

A common method for ensuring that two transceivers have an unobstructed view of one another is found by estimating the dimensions of the first Fresnel zone. This zone corresponds to an ellipsoid where the major axis aligns with the straight line connecting the transmit and receive antennas (Fig. 11). Blockages or flat surfaces existing inside the first zone (ellipsoid) may offer a secondary path for transmitted energy to be received out of phase with the primary transmission. As this particular BVLOS flight took place over open water (ideal reflector), it is critical to ensure that the ellipsoid encompassing the ground and aircraft antennas do not to intersect the physical ground, or other objects extending above the surface (trees, buildings, etc.).

Table 4 Fresnel minimum altitude limits for C2 links

C2 System	Frequency	Minimum altitude at max radio range (Table 3)	Minimum altitude at max range for course flown($d = 13.5\ nm$)
Telemetry	915 MHz	4633 ft	252 ft
Control	433 MHz	4408 ft	259 ft
ADS-B	978 MHz	75,653 ft	251 ft
Video	5.8 GHz	39 ft	280 ft

Fig. 12 Photo taken on outbound leg of record flight. Failed nose gear is clearly visible and the impact to the flight is detailed in section 4.1

The expression for the Fresnel radius is given by [26].

$$F=\sqrt{\frac{n\lambda d_1 d_2}{d_1+d_2}}, \tag{3}$$

where $d_{1,\ 2}$ refers to the distance from the transmitting and receiving antennas respectively, λ is the wavelength of interest, and n is the zone number of interest. The maximum (most conservative) Fresnel radius for the first zone occurs at $d_1 = d_2 = d$ giving,

$$F=\sqrt{\frac{\lambda d}{2}}. \tag{4}$$

Table 3 lists the Fresnel radii for the C2 radios used on the Avanti.

GCS antennas were affixed to a tracking gimbal positioned 6 ft. above ground level (AGL). The minimum flight altitude must lie outside the first Fresnel zone, and requires estimating the continuously computed ellipsoid dimensions between ground and vehicle antennas,

$$h_r = 2F - h_t + \frac{d^2}{R_E k}, \tag{5}$$

where $h_{t,\ r}$ are the height of the transmit and receive antennas, and R_E ($6{,}371 \times 10^3\ m$) appearing in the denominator of the

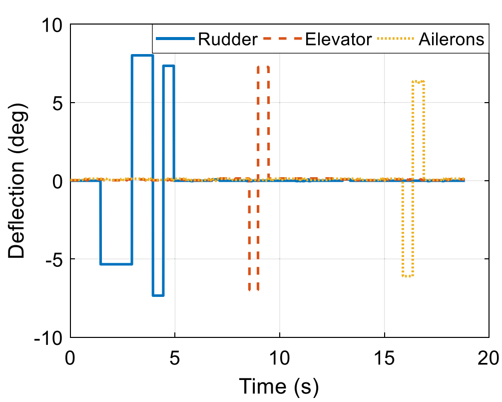


Fig. 13 Control surface input history for system identification flight

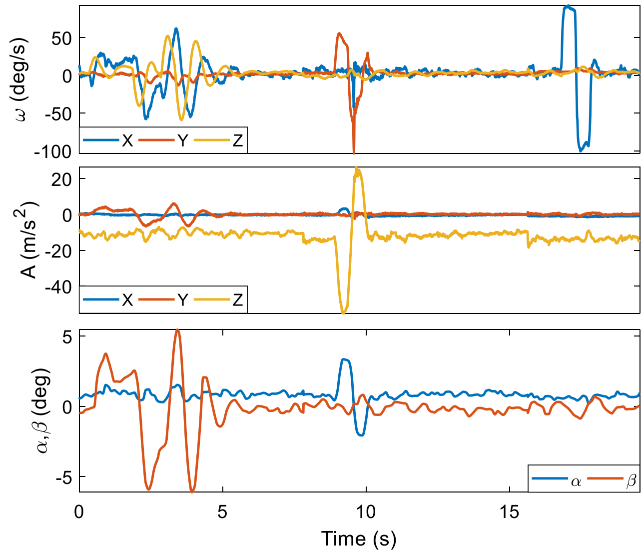


Fig. 14 Time history of measured vehicle response during system identification flight

rightmost term represents the intersection of the curved Earth surface with the first Fresnel zone. The atmospheric propagation constant, k is approximately 1.33 (air), and the distance, d between transmit and receive antennas according to the maximum range figures outlined in Table 3. With the transmit antennas located at 6 ft. AGL, the minimum altitude for each C2 link is listed in Table 4. In this table, the minimum altitude at the maximum reported range of the radios is shown. This altitude corresponds to the height at which the aircraft must fly to ensure there are no encroachments into the first Fresnel zone. The chosen base of operations and location of the GCS was made to ensure clear line of sight with a maximum distance from transmit to receive antennas of 13.5 nm. When operating within this distance of the GCS, the aircraft must be operated above 260 ft. AGL to remain outside the first Fresnel zone. Beyond the airport environment, the Avanti was operated at a cruise altitude of 1000 ft. AGL, thus ensuring adequate C2 availability.

4 Flight Testing

Flight testing data for this work came from three flights leading up to the BVLOS mission outlined in section 2.1. Due to weight constraints encountered with the additional fuel required for the BVOLS mission, and the need to keep the maximum takeoff weight fixed (FAA approval requirements), ballast weight was removed from the nose of the aircraft and the CG location recomputed. Initially the aircraft had been operating with a static margin of approximately 15%. The removal of the nose ballast pushed the static margin to 5%, a noticeable change. However, the flight guidance laws were based around real-time system identification techniques

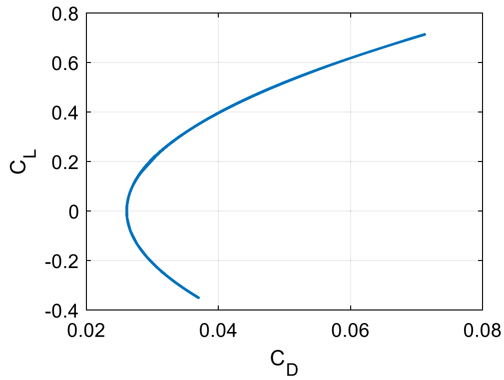


Fig. 15 Lift and drag coefficients for the Avanti in a clean configuration

developed in support of the record flight [27, 28]. Within approximately five seconds of leaving the ground, the aircraft had converged on a new set of guidance gains sufficient to ensure that the aircraft was stable and controllable within the predicted flight envelope. Previous flight-testing had established a baseline flight model using a combination of multi-sine orthogonal inputs, doublets, and 3–2–1-1 control surface excitations [7, 29, 30]. An adaptive LQR and pseudo-dynamic inversion technique was used for control [31–35]. During the BVLOS flight, the nose gear failed to retract, providing an interesting opportunity to prove the validity of the system identification methods (Fig. 12) and is detailed in the following section.

4.1 Aircraft Range Measurement

Low-altitude aircraft range and endurance estimates are primarily dependent upon the TSFC and *L*/*D* characteristics of the vehicle. Therefore, characterization of these quantities is of prime importance to ensuring safe BVLOS operations. Limited TSFC information is provided by the engine manufacturer but does not include performance reductions due to installation effects. To compute an installed TSFC through flight test, the vehicle *L*/*D* characteristics first need to be identified. The speed of the aircraft made classical flight tests for estimating the drag polar difficult given the requirement to remain within visual contact with the vehicle, and below 400 ft. Therefore, the first flight tests used an output-error system identification methodology to estimate the aerodynamic characteristics of the aircraft. However, another data source for estimating the drag polar occurred when transitioning out of the high-speed mission segments where the engine was placed at idle to conserve fuel. The resulting level deceleration allowed for direct assessment of the drag polar and will be presented for comparison with the system identification techniques.

The system identification flights performed used an output-error filter framework with a model of the lateral and longitudinal dynamics of the aircraft to provide refined vehicle stability estimates and the aerodynamic characteristics necessary to fly the aircraft BVLOS. Briefly, system identification is a means of identifying a six-degree of freedom mathematical model (pseudo transfer function) for the aircraft under test representing its response to control inputs from all available flight control surfaces. This methodology has proven successful at identifying linear and non-linear flight models for a wide variety of flight vehicles [27, 36]. To identify a model with sufficient accuracy for control law development and range estimation, all control surfaces must be actuated in such a manner that the resulting vehicle response (output) is clearly correlated to the commanded input across an appropriate excitation frequency range. Correlation can be reinforced using orthogonal control inputs generated by phase shifted sine waves, or other pre-processed orthogonal polynomial or wavelet sets, however satisfactory results have been achieved in this work using simple doublets, and 3–2–1-1 excitations (Fig. 13) with the vehicle response shown in Fig. 14. For this work, the expected short and long-period modes common to most aircraft were expected to occur at higher frequencies when compared to full-scale aircraft. Therefore, the doublet and 3–2–1-1 excitations were scaled to provide more excitation energy at higher frequencies (shorter actuation period).

The input data for the system identification flight requires accurate knowledge of the angular deflection of the vehicle's moveable control surfaces. This was achieved by correlating the commanded servo position with the resulting surface angular deflection. The servos chosen for this aircraft provided more than five times the required torque expected during the most aggressive maneuvering, ensuring the commanded and actual positions were equivalent during flight. Additional

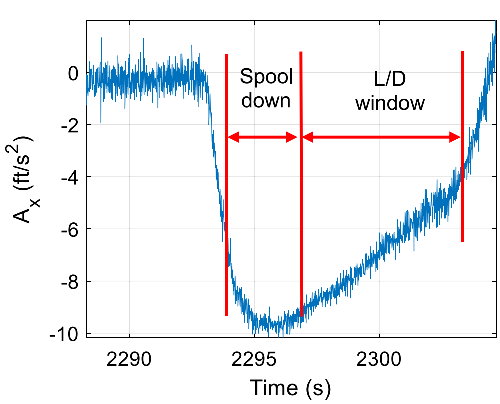


Fig. 16 Measured vehicle longitudinal acceleration with a spool down (occurring near 2295 s) reducing the thrust to approximately zero pounds

Table 5 Drag and efficiency factors for clean and nose gear down configurations

Configuration	C_{D0}	e
Clean	0.026	0.72
Nose Gear Down	0.033	0.68

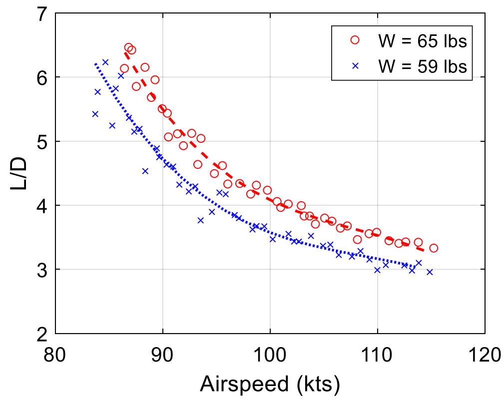


Fig. 17 *L/D* estimated during outbound (red) and inbound (blue) legs

measurements included the aerodynamic flow angles (α, β), the vehicle body-axis accelerations, angular rates, and velocity/thrust estimates. Aircraft geometric data (wing area, span, etc.) were measured prior to flight, and inertial parameters were estimated using a basic CAD model of the airframe with point masses representing major internal components.

The aircraft models used for later-longitudinal motion included a linear lift model and a parabolic drag model,

$$\begin{aligned} C_L &= C_{L_0} + C_{L_\alpha}\alpha \\ C_D &= C_{D_0} + \frac{C_L^2}{\pi e AR} \end{aligned} \tag{6}$$

where C_L is the vehicle lift coefficient, C_{L_0} is the lift coefficient at zero angle-of-attack, C_{L_α} is the lift curve slope, α is the angle of attack, C_D and C_{D_0} are the vehicle drag and drag and zero-lift coefficients, e is the Oswald efficiency factor, and AR is the wing aspect ratio.

These models were used as the basis for identifying the unknown lift and drag coefficients and the Oswald efficiency factor, e. Additional models were used to identify stability coefficients, but are not presented in this work. The output-error modeling methodology is akin to a non-linear least-squares best-fit solution which minimizes the error between measured and modeled response characteristics. Confidence intervals for the resulting model are simultaneously computed

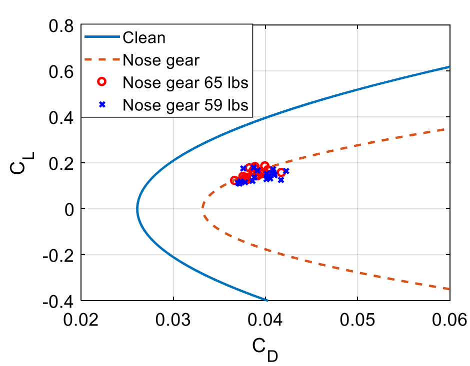


Fig. 18 Lift and drag polar for Avanti in the clean configuration and with nose wheel down

to ensure that the estimates are uncorrelated and correctly identified. The resulting drag polar estimate for the Avanti in a wheels-up, clean configuration (derived from system identification flight testing) is shown in Fig. 15, with $C_{D_0} = 0.026$ and $e = 0.72$.

The vehicle *L/D* was estimated in flight by commanding level-flight decelerations such as occurs preceding a turn and when leaving the high-speed legs. During these mission segments, the vehicle throttles down to preserve fuel, from a high-speed dash of ~140 kts down to 100 kts for cruise. During the deceleration, the turbine decreases thrust to idle, which in static testing corresponds to approximately 0 lbs. of thrust (a sensitivity analysis shows limited impact of this assumption). During the deceleration, the altitude controller ensures the vehicle remains close to its assigned altitude ±1.5 ft. A simple balance of the Earth-frame acceleration and drag force acting in opposition to the motion of the vehicle results in a simple expression for estimating drag as a function of velocity,

$$\sum m a_x = -D = -C_D q S \tag{7}$$

where m is the instantaneous vehicle mass, a_x is the acceleration in the vehicle lateral axis, D is the overall vehicle drag in lbs., q is the dynamic pressure, and S is wing area.

Estimating C_D as a function of vehicle velocity requires only the Earth-referenced longitudinal acceleration, airspeed, static pressure, and temperature under the assumption of zero-net thrust. The air-data system used in the aircraft provides these measurements from pressure and temperature sensors embedded in the five-hole probe extending from the nose of the vehicle. Body-referenced accelerations provided by the strap-down IMUs can be transferred to the Earth reference frame using the estimated state quaternion. The translated accelerations during the deceleration maneuver are shown in Fig. 16 where the regions corresponding to idle thrust are clearly visible. Estimates of the lift produced were simply computed as the gross takeoff weight minus the accumulated fuel burn (provided by the turbine engine control unit).

The computed *L/D* ratio is shown in Fig. 17 for the deceleration windows experienced following the first and second high-speed runs where the vehicle weight is reduced from 65 lbs. to 59 lbs. From this data, a second-order surface fit was used to parameterize *L/D* as a function of vehicle velocity and weight, allowing for estimation of the installed thrust during the high-speed phases of flight. For the unique case of flight with an extended nose gear, the resulting zero-lift drag coefficient was ~0.033 (summarized in Table 5), quite high compared to most modern jet aircraft. Drag buildup methods proposed by Horner [22] and Raymer [23] specify that the extended landing gear leg (un-faired strut and wheel, 5.6 counts) and exposed gear bay (1.4 counts) added approximately 7 drag counts, and reduced the maximum attainable velocity from 138 kts to 124 kts. When the system identification

Table 6 Measured mission segment weight fractions

Segment	W_f/W_o *Measured*	W_f/W_o *Estimated*	*Measured Range*	*V*	*Measured Lift/Drag*	TSFC (1/h)
Engine start/taxi/departure	0.990	0.970*	–	–	–	–
Climb	0.981	0.998	2.2 nm	86 kts	8.3	5.58
Gate 1	0.986	0.992	2.7 nm	96 kts	4.6	2.53
High-Speed 1	0.954	0.979	8.1 nm	128 kts	3.1	2.18
Gate 2, 3, & Turnaround	0.958	0.982	7.3 nm	94 kts	4.4	2.40
High-Speed 2	0.943	0.979	8.1 nm	127 kts	3.0	2.70
Gate 4	0.985	0.991	2.3 nm	94 kts	3.8	2.30
Landing	0.996	0.995*	–	–	–	–
Total	0.813	0.894				

methods were applied to the vehicle with the extended landing gear leg, the resulting *L*/*D* curve clearly shows the increased drag associated with the unusual configuration (Fig. 18). In this figure, the data points correspond to the *L*/*D* estimates made during the deceleration maneuvers at different weights.

Measured and estimated weight-fractions for each mission segment, with range, airspeed, and *L*/*D* are shown in Table 6. The Breguet range equation allows for the estimation of the TSFC using the available flight data. Of note, the Breguet assumption of no-wind conditions was not applicable during the record flight as evidenced by the differing TSFC estimates (2.18/h during leg 1, and 2.7/h during leg 2) in the high-speed runs. The average TSFC (2.44/h) is considerably higher than full-scale turbojet powerplants, but within the manufacturer's specifications of 2.34/h used in the initial range estimation. The total fuel burn for the mission was 1.92 gal, leaving 1.25 gal in reserve, compared with 1.09 gal estimated for the flight. The discrepancy in the predicted and actual fuel consumption is due to the higher than predicted TSFC in conjunction with the increased drag due to the extended landing gear and correspondingly lower *L*/*D*.

As shown in Table 6, the Engine start/taxi/departure used less fuel than anticipated due to starting on the runway, rather than taxing out. The hold-time on the ground to verify system functionality was also less than anticipated, resulting in a lower overall fuel burn. The climb phase used more fuel due to the increase drag of the landing gear, resulting in a decreased climb rate at a lower airspeed. As noted, the remaining legs used more fuel due to the increase in TSFC and the increased drag of the extended landing gear. Operating in an off-nominal condition can dramatically affect the range and endurance of already fuel-limited small UAS and should be of first-order importance in developing fuel minimums and real-time performance estimation techniques for BVLOS missions. Of note, if real-time estimates of the vehicle drag polar are available, the range equation models noted in section 3.1 can be iteratively updated to provide refined estimates of the range and endurance.

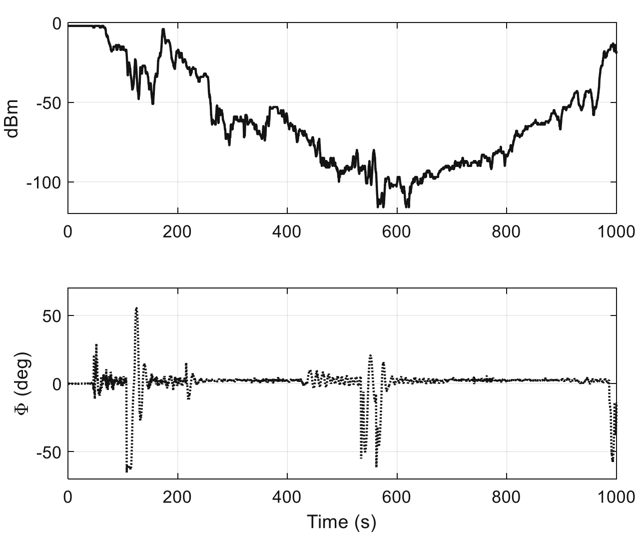


Fig. 19 Received power showing insensitivity to vehicle roll attitude for 433 MHz radio

4.2 RF Range Measurement

This section details the estimation of the RF signal propagation characteristics through flight testing, to confirm the initial design estimates and validate the propagation models

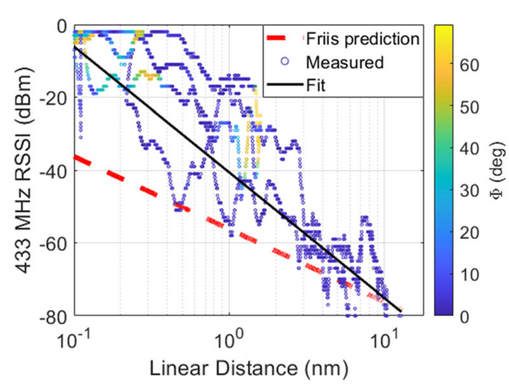


Fig. 20 RF receive power measured in flight for 433 MHz radio

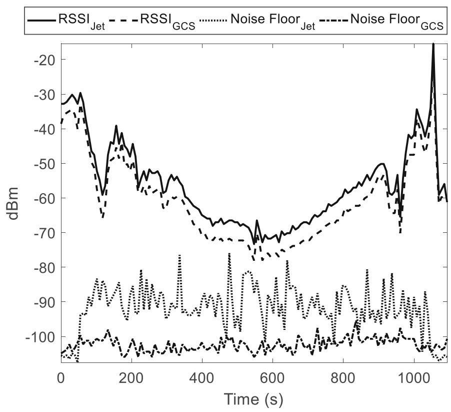


Fig. 21 Received power data for 915 MHz radio. GCS refers to the ground control station telemetry radios, and jet refers to radios on board the aircraft

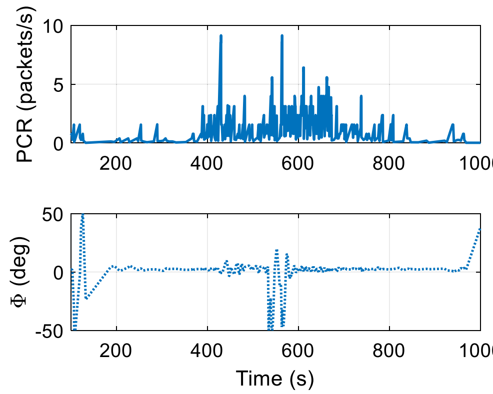


Fig. 23 Packet correction rate (PCR) (top) not strongly correlated to bank angle at maximum range (bottom)

presented previously. Of note, not all radio systems onboard reported their transmit power levels or indications of received signal strength. The RDF900+ 915 MHz radio provided received signal strength indication (RSSI) information, which allows for computation of the actual received power using calibration relations provided by the manufacturer. The RDF900+ radio also provided noise floor data which allowed for a real-time analysis of the maximum range using a given power output level. Similarly, the DragonLink V3 433 MHz link provided RSSI and noise floor information. The satellite communication system did not report received signal strength on the aircraft, however, ground monitoring was possible using a second receiver co-located with the operators. Unfortunately, the satellite data link dropped when the aircraft was approximately 5.4 nm from the takeoff location, the cause of which is still under investigation. It is envisioned that this system will be the primary link for future operations given that it does not require horizontal line-of-sight between the ground control station and the aircraft. However, the antenna it uses is highly directional to facilitate communications with a geostationary satellite and thus requires gimbaling on the aircraft, or limitations on the pitch/roll excursions of the vehicle. The 978 MHz ADS-B radio (Ping2020) did not provide a direct

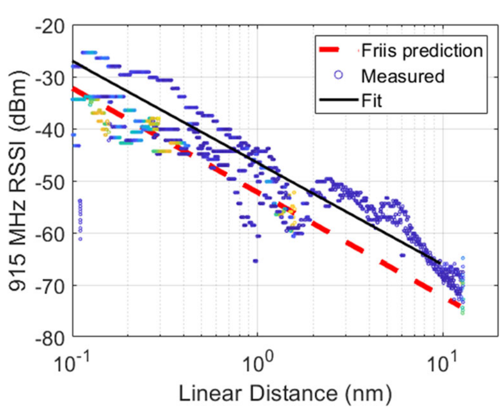


Fig. 22 RF receive power measured in flight for 915 MHz radio

estimation of received signal strength, making flight analysis impossible. Similarly, the 5800 MHz link did not provide RSSI or noise floor information and was not used during the record flight as the limited visual information was not deemed critical to mission success. This remainder of this section details the analysis of the antenna installations in the aircraft, and corroborating measurements made during the record attempt for RF link budget analysis.

Flight-test data was gathered for the 433 and 915 MHz systems to assess the Friis model as a range prediction tool for BVLOS missions. In this testing, the aircraft and ground-based transceivers were set to the maximum output power of 1 W. For the data presented below, the receiving antennas were located on a 6 ft. ground station tower with co-located receiving antennas for the 433, 915, 978, and 5800 MHz antennas. The ground tower featured a custom tracking head which would orient the directional ground antennas to point directly at the vehicle as it traversed the course.

The measured signal strength for the 433 MHz system is shown in Fig. 19 alongside the roll angle, Φ. The bank angles during flight frequently approached 60° to stay within the relatively narrow boundaries of the course. Though as shown in Fig. 19, there is little correlation between the bank and received power levels for the radios considered in this work. For the mono and dipole antennas used on the communications radios, this drastically reduces the gain of the system by effectively inducing a polarization change. The resulting range versus signal strength plot is shown in Fig. 20. In this plot, the Friis predicted signal strength is shown as a red dashed line, and the curve fit for RSSI shown as a black line. The error between the mean RSSI and Friis predictions is likely traceable to higher power output levels (5–10%) from the radio transceivers. The signal fades closely follow the Friis attenuation bound when the signal strength drops due to multipath and roll effects demonstrating the effectiveness of the simple Friis model when adhering to the assumptions outlined in Section 3.2.2.

The 915 MHz radio receive power and noise floor data are shown in Fig. 21. Of interest, the noise floor from the receiver on the aircraft jumps by 20 dBm at around 50 s, corresponding to start of the turbine. This was traced to radiated noise from the unshielded turbine spark-ignition system. Proper grounding of high-powered electronics in a small-scale composite airframe is a formidable challenge. A major takeaway from this work is the recommendation that future vehicles should incorporate proper grounding allowances in the initial design of the airframe, rather than retrofit through testing. Whereas the signal attenuation for the 433 MHz link is somewhat obscured by the error checking and mitigation routines built into the radio, the 915 MHz link clearly shows the applicability of the Friis based range prediction algorithm (Fig. 22). Of note, the RSSI levels for the 915 MHz radio are slightly higher than the Friis predictions due a slightly higher power output of the radios (on the order of 10–15%) and should be considered when relying on reported power output specifications. The 915 MHz radio also features a diversity receiver and error correction algorithms which utilizes additional bandwidth to resend and/or correct corrupted packets. Figure 23 demonstrates the effect of the error correction as the aircraft approaches the maximum distance from the GCS at approximately 500 s. At this point, the telemetry receiver is correcting approximately 3 packets per second (approximately 6%) due to signal drops and fades. The diversity component of the receiver helps to correct for fades due to the rolling of the aircraft (effectively changing the polarization of a single antenna). In the same figure, the roll angle is plotted to demonstrate the insensitivity of the packet correction rate (PCR) to the effective polarization of the diversity antennas, a key feature for link integrity when approaching the limits of the systems range.

5 Conclusions

This work addresses the critical design considerations for developing a high-performance BVLOS vehicle. While many of the testing methods and analysis tools have been applied to full-scale vehicles, their applicability to small UAS platforms has remained circumspect. To demonstrate these principles, a mission was designed around the FAI autonomous speed and distance records for a small UAS. A notional vehicle was chosen and the procedures for assessing its ability to meet mission requirements was outlined, focusing on range and endurance estimation as well as propulsion system characterization. Estimation of unknown aircraft performance data such as the drag polar and *L/D* as a function of vehicle weight were demonstrated and shown to have a significant influence on the resulting maximum speed and range. This work underpins the necessity of including such analyses in the initial mission planning phases for BVLOS flights, and points to the need for continued development of lightweight, real-time performance estimation techniques for UAS. Additionally, this work demonstrates the practical flight-testing techniques which can provide such performance data for existing air vehicles with little to no supporting documentation.

While vehicle performance is one aspect of demonstrating safe BVLOS flight, a robust command and control link is another critical component unique to UAS operations. To this end, we developed range prediction tools based on the Friis transmission relation for five independent vehicle links. These included direct RF line-of-sight systems operating in the UHF bands (433, 915, and 978 MHz), one L-band satellite communications link (1.5 GHz), and one C-band video link (5.8 GHz). In addition to range estimation, Fresnel limits were identified which established minimum safe altitudes to ensure a robust communication link over the course. The range prediction algorithms required precise knowledge of the gain parameters for the aircraft, so each system was characterized in a compact range facility at the Ohio State. These range estimates were validated in the BVLOS mission by recording received power levels and noise-floor data for several of the RF links. The flight data and Friis attenuation predictions compared favorably for the 433 and 915 MHz links. Polarization issues were identified by inspection of the aircraft roll angle compared to packet drop rates and power attenuation with little noted correlation. Additionally, use of multiple high-powered RF systems in relatively close proximity encased within a composite airframe posed significant challenges from an integration and testing perspective. The lack of an adequate grounding system in the airframe resulted in numerous ground loops, EMI, and signal fade issues which had to be addressed before operating BVLOS. Considering these challenges, integration of the communication and control links into the airframe should be prioritized on the same level as the flight control system and powerplant selection as it is a critical design constraint somewhat unique to UAS vehicles.

Systems level descriptions and measurement capabilities were presented for the inertial measurement system and flight management computers. These systems provided the real-time data for controlling the vehicle during the mission, and high-fidelity data for post-flight analysis. Details on the strap-down mechanization unique to the high-speed flight of small UAS were discussed and the influence of external aiding sensors on navigation performance detailed. This work utilized a differential RTK GPS system to correct the errors introduced by uncompensated bias shifts in the IMU, though the usefulness of these corrections diminishes as the distance from the ground-based fixed receiver increases. The corrected state information provided through fusion of these sensors was used in conjunction with direct measurement and system identification techniques to assess the drag characteristics of the vehicle. During the BVLOS mission, the nose gear failed to retract, providing a unique opportunity to assess the fidelity of the drag prediction models and the resulting impact on range and endurance of such a vehicle. This incident demonstrated the first-order importance of monitoring the vehicle performance in flight as the drag increase significantly impacted the maximum speed and range of the vehicle. Lessons learned from this point to the necessity of incorporating real-time health and performance monitoring capabilities for high-performance vehicles operation BVLOS. Ultimately, this vehicle was accepted for the official FAI speed record over a 8–13.5 nm course for Class U-2.b, group I UAS at 147.20 mph, and for the an out-and-back distance record on the same course of 24.4 nm.

Acknowledgments The authors wish to thank Ligado Networks (Mike Gagne, Sam Welch, David Nance, and Illya Ziskind) for their generous donations in time and resources for the satellite communication system. Additionally, the students of the Aerospace Research Center UAS Laboratory at the Ohio State University, Ross Heidersbach, Rokhaya Diawara, Madhav Shah, and Braxton Harter, assisted with aircraft fabrication and renovation. The ground crew during flight operations was Daniel Applebaum, Ryan Thorpe, Mark Sutkowy, and Wenbo Zhu. And finally, we wish to thank the chase plane crew of Ryan Smith, Jamie Struewing, and Ross Heidersbach.

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